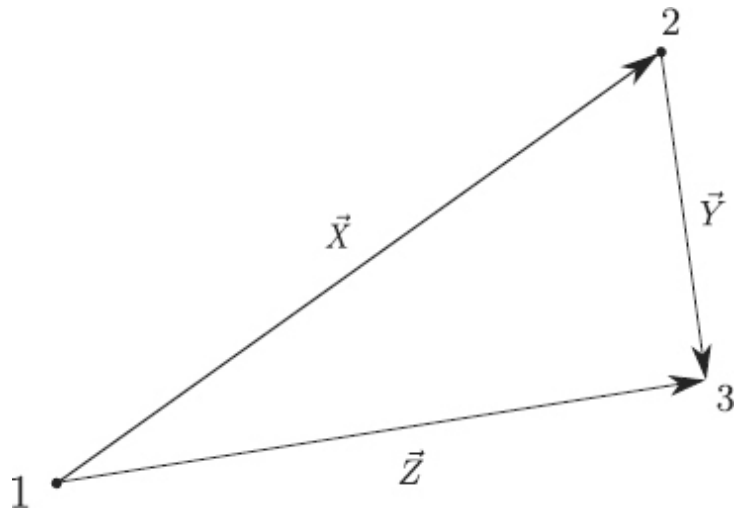


Figure 5.1: The Triangle Inequality. The sum of the lengths of vectors $\vec{X}$ and $\vec{Y}$ is greater than or equal to the length of vector $\vec{Z}$. (The shortest path between two points is a straight line.)

The triangle inequality can be expressed in more than one way. We'll start with the basic definition and then massage it into the form we need. We know that

$$|X| + |Y| \geq |Z|.$$

If we think of *X* and *Y* as vectors that can be added, we can write the above as

$$|\vec{X}| + |\vec{Y}| \geq |\vec{X} + \vec{Y}|.$$

If we square this equation, it becomes

$$|\vec{X}|^2 + |\vec{Y}|^2 + 2|\vec{X}||\vec{Y}| \geq |\vec{X} + \vec{Y}|^2.$$

But the right-hand side can be expanded as

$$|\vec{X} + \vec{Y}|^2 = |\vec{X}|^2 + |\vec{Y}|^2 + 2(\vec{X} \cdot \vec{Y}).$$